PROJECT TITLE

Climate Classification System using Machine Learning

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Introduction

This project uses Machine Learning to identify the season from weather data. It takes temperature, humidity, wind speed and rainfall as inputs. A trained Random Forest Classifier predicts whether the weather belongs to Summer, Monsoon or Winter.

Project Overview

The project performs the following steps:

- Stores six sample weather records with their correct season labels.
- Uses four weather features: temperature, humidity, wind speed and rainfall.
- Assigns numeric labels: 0 for Summer, 1 for Monsoon and 2 for Winter.
- Trains a Random Forest machine learning model using the sample data.
- Tests the trained model with three new weather examples.
- Displays the predicted season as readable text.

Features

- Accepts four weather measurements as input.
- Classifies weather into three season categories.
- Uses 100 decision trees in the Random Forest model.
- Predicts the climate category for unseen input data.
- Displays the result with season names and simple weather details.

Training Data

Temperature (°C)	Humidity (%)	Wind (km/h)	Rainfall (mm)	Season
38	40	15	0	Summer
35	45	12	2	Summer
27	88	25	45	Monsoon
25	92	30	60	Monsoon
15	60	10	0	Winter

Temperature (°C)	Humidity (%)	Wind (km/h)	Rainfall (mm)	Season
12	55	8	0	Winter

Technology Stack

Component	Technology / Purpose
Programming Language	Python
Numerical Library	NumPy (imported in the program)
Machine Learning Library	Scikit-learn
ML Algorithm	Random Forest Classifier
Learning Type	Supervised Machine Learning

How the Program Works

- **Import libraries:** imports NumPy and RandomForestClassifier.
- **Prepare data:** X contains the four weather inputs and y contains the correct season numbers.
- **Map labels:** climate_names converts 0, 1 and 2 into Summer, Monsoon and Winter names.
- **Create the model:** RandomForestClassifier(n_estimators=100, random_state=42) creates the classifier.
- **Train the model:** model.fit(X, y) teaches the model using the six labelled examples.
- **Test new data:** model.predict(new_weather_data) predicts the seasons for three new examples.
- **Show output:** a loop prints each weather record and its predicted climate name.

Test Cases

Test	Temperature	Humidity	Wind	Rainfall	Expected category
1	36°C	42%	14 km/h	0 mm	Summer
2	26°C	90%	28 km/h	50 mm	Monsoon
3	14°C	52%	9 km/h	1 mm	Winter

Software Requirements

Python environment with scikit-learn and NumPy installed. The program can be run in a Python IDE or notebook environment such as Visual Studio Code, Jupyter Notebook or Google Colab.

Conclusion

This project demonstrates how a supervised Machine Learning model can learn from labelled weather examples and classify new weather measurements into Summer, Monsoon or Winter.

Note: The example uses only six training records. It is a basic learning demonstration, not a reliable real-world weather forecast. More real and varied weather data would be needed for dependable predictions.